

PIR - Inflated allocations at Thatcham depot

M&S Digital&Tech Post Incident Review



Incident/ Problem Reference: 90610998

Incident Date: 18/02/2025

PIR Date: 21/02/2025

Author: Saloni Chowdhury

Technical Owner: **Andrew Goslin & William Hanson**

PIR Reviewer: **Richard Beaumont & William Hanson**

1. Incident Summary :

Provide a concise overview of the incident with brief update on issue, impact, root cause and resolution.

On 17/02, a new functionality was tested by GIST from Thatcham RDC for Hedge End Store, leading to stock corruption due to a mismatch in the "into store delivery date" between the product and DWS system. Multiple adjustments were made in CSSM to mitigate the impact. Subsequently, on 18/02, GIST reported inflated allocations at Thatcham Depot and contingency allocations were applied to mitigate impact to the stock deliveries across 150 stores serviced by Thatcham. The root cause was attributed to an issue within CSSM leading to the system not accounting for the adjustments on 17/02 thus sending negative stock in transit values to RELEX. Services were restored by deploying a fix in CSSM and clearing down the SAND (stock allocated not dispatched) values in ASO.

On 22/02 - Retail ops highlighted inflated allocations and deliveries to Hedge end store due to the negative stock value in CSSM - RTR (Real time response) API where CSSM could not handle the negative value due to the similar issue in CSSM. To mitigate the impact, the same fix used for EOD file on 18/02 was implemented on RTR API as well and the UPC's (~10) were handled internally to prevent any wastage.

2. Incident Chronology :

Include a record of the data/time of key events. e.g. "07:36 Incident started, notified by..." **keep fact based**.

17/02:

Within DWS (Depot Warehousing System) v76 there is now a function to transfer picked stock from one store pallet to another store. This was actioned at Thatcham at circa 07:45GMT this morning (12 trays from Brooklands store to Hedge End -1643 store) BUT this has culminated in positive stock adjustment messages of 18,096 trays impacting ~2490 UPCs.

No impact to Brooklands store. There are stock adjustment errors (1185 - 4 messages) and goods receipt errors due to (1181 - 137 messages for multiple suppliers) messages.

Store impact - **2,490** stock positions were corrupted with +171,735 artificial stock added. Store has counted (in gap scan/accuracy check process) **39** UPCs were counted by store this morning creating a **loss** of approx £10,000 - Retail Ops were contacted by the CSSM team.

CSSM team confirmed that the stock adjustments had corrected the stock position for Hedge End store and that the data for the 39 UPCs which

were counted was corrected.

DWS planning to revert v76.05 code change in Thatcham (no other depots impacted). Menu option has been revoked in the meantime

18/02 :

This morning, GIST has reported inflated allocations generated for Hedge End store impacting the stock movement from Thatcham depot across the stores. Therefore, contingency allocations were applied for all depots at 09:29 and GIST confirmed that the allocations are now being generated as expected for Thatcham.

At 09:59, all depots except Thatcham were put back to live allocation. We are not anticipating any further impact on stores today. Potential impact to Thatcham operation as they might receive a larger than expected volume of deliveries to the stores.

Deep dive into the cause behind the inflated allocations this morning - during the recovery of yesterday's stock position for Hedge End , some part of adjustments might have been overlooked which could have caused the issue.

Actions are being performed to correct the stock files across ASO and CSSM to generate correct allocations for tonight's batch.

22:30:

1. The bug fix has been deployed in CSSM and validations have confirmed that there are no negative values within the SIT (Stock in Transit) file from CSSM.
2. Also, validations were performed on the overall SIT value no abnormalities observed.
3. As agreed, ASO team has zeroed the SAND (Stock allocation and not dispatched) value for the 632 UPCs /1904 trays, after taking the required backup. *Note:* Out of 3392 trays that were in the original list for 632 UPC's,1488 has already been dispatched thus the remaining 1904 trays were zeroed to ensure that there are no further avoidable future ordering / allocation anomalies. - Validations from SAND, were also positive no abnormalities observed.
4. Once the CSSM SIT extract is completed at 23:00, transfer jobs will be released and a final validation will be performed by Gayam, Mrunalini/FOAL team before loading the file into RELEX.

Hyper care monitoring is in place across all the systems overnight

.

3. Incident Resolution

 *The actions taken to restore services and mitigate impact.*

On 17/02, multiple adjustments were made in CSSM to recover the stock position for Hedge End Store (1643). Anomalous good receipts (which had failed systematically) were not sent to subscribers.

On 18/02, contingency allocations were applied at Thatcham to mitigate the impact on stock deliveries across stores. A permanent bug fix was deployed in CSSM, and SAND (stock allocated not dispatched) values were cleared for Hedge End Store allocations. Additionally, Thatcham Depot performed manual adjustments and allocations to support recovery.

4. Root Cause

 *Refers to the fundamental reason (s) why an incident occurred including problem description, analysis techniques, contributing factors, evidence with concise root cause statements*

The issue was caused by a combination of factors:

1. The code used by GIST in their new capability to transfer picked stock between store pallets revealed an issue due to a mismatch between the into-store delivery date of a product (18/02) and the corresponding date in DWS (17/02), generating a large volume of positive stock adjustment messages. The scenario had not been covered by the GIST QA team.
2. In CSSM, negative adjustments were made to correct store stock positions. However, due to an existing issue with the CSSM logic not accounting for the negative DC adjustments (which are not receipted by systematically), this resulted in negative stock-in-transit values being sent to RELEX.

Note: Quantum was able to handle these negative values by treating them as zero. However, due to an underlying behavioural issue in RELEX, these values were incorrectly processed. An EPIC has been raised with RELEX to investigate why it is considering negative stock-in-transit values

Context:

GIST - A new function in DWS version 76 allows picked stock to be transferred between store pallets to reduce manual work at depots. This was first actioned at Thatcham at 7:45 AM, where 12 trays were moved from Brookland to Hedge End store. However, this triggered 18,000 positive stock adjustment messages, impacting 2,400 UPCs. The issue stemmed from outdated 2011-era code used in the capability. **While recreating the scenario in a lower environment, it was found that a mismatch between the into-store delivery dates (DWS showing the 17th, actual date being the 18th) led to multiple positive adjustments being sent to CSSM, causing stock corruption.**

Currently, this problematic code is only in Thatcham (version 76.05), while all other RDCs remain on version 76.04. **Discussions are ongoing with**

the GIST CIO on whether to fix forward or remove the code, which would require GIST to continue manual store transfers.

Note: GIST has disabled code v76.05 in DWS, however, not fully reverted

GIST confirmed that the capability underwent multiple test cycles, and no issues were found in DWS lower environments. However, there were no communications to M&S about the live test on Monday. As a corrective action, GIST has agreed to prioritise communication on any future smoke tests to align on proactive hypercare and avoid business impact.

CSSM - On 17/02, after identifying the issue, all relevant teams—including Foods, SAP, and CSSM—conducted thorough investigations across multiple interfaces, including adjustments, GRs, ASNs, STOs, and POs. To mitigate the impact, several reversions were performed in CSSM against artificial stock gains at the stores. However, since stores had already conducted stock counting post-GIST's test, this affected 39 UPCs. CSSM manually updated these UPCs to align stock levels.

On 18/02, validations confirmed that the CSSM stock file had negative values due to existing CSSM logic that does not factor in unreceipted (and negative) DC adjustments, negative stock-in-transit values were sent to RELEX, leading to over-allocations. **Previously, Quantum could adjust these negative values to zero, ensuring proper allocations, but due to an issue within RELEX, this is no longer accounted for. An EPIC has been raised for RELEX to investigate and resolve the issue. We have agreed an action to improve the scopes of hypercare within CSSM/ASO to identify the negative value prior to it being sent to Relex during such a scenario.**

To correct stock positions, an alternative reversal approach is being assessed within CSSM, avoiding the creation of new events through negative adjustments.

Key CSSM Interfaces Involved:

- I-185 (Food Allocation Backup Route): Auto-acknowledges additional positive adjustments triggered by GIST, updating stock levels.
- I-185 DC Adjustment Backup Route: Reverts food allocation messages and corrects adjustments triggered by GIST via negative transactions.
- I-3451 (End-of-Day Free Stock-in-Transit File): Sends stock-in-transit data to RELEX. CSSM's logic did not previously account for DC adjustments, leading to negative values being sent.

On 18/02, a fix was applied in CSSM to ensure DC adjustments are considered and prevent negative stock-in-transit values from being sent to RELEX.

5. Findings

5.1. Observability –

On 17/02, GIST highlighted the issue when their capability testing failed at Thatcham for Hedge End Store. We have requested GIST to provide full visibility into their monitoring mechanisms to detect threshold breaches in adjustment messages and to adhere to D&T change management process for their future capability tests. Additionally, we explored whether inflated allocations in RELEX could be identified at an individual store level. However, it has been confirmed that the existing monitoring only tracks inflated allocations across all stores, making it impossible to isolate over-allocations for a single store. As a next step, GIST has agreed to review their monitoring capabilities and assess whether an issue can be detected at either a single-store or multi-store level prior to the allocation being generated.

5.2. Technical Challenges -

Due to a lack of prior communication, the recovery actions were initially reactive. However, once the issue was identified, the product teams collaborated effectively to reverse the positive adjustments in CSSM.

Moving forward, we are exploring an alternative approach in CSSM to handle both positive and negative adjustments without creating additional events with random data. This will ensure a technically sound recovery process while preventing cascading impacts on other systems. Further collaboration with SAP and downstream systems is required to validate this approach, as these systems receive adjustment messages.

Also, Previously, Quantum could adjust these negative values to zero, ensuring proper allocations, but due to an issue within RELEX, this is no longer accounted for. An EPIC has been raised for RELEX to investigate and resolve the issue

There is also an opportunity to enhance hyper-care validations after the adjustments are performed in CSSM, ensuring that all interface files—including CSSM, ASO are validated to prevent unintended impacts.

5.3. MIM involvement & impact assessment -

On 17/02, the issue was reported to MIM, but since the impact was limited to one store, the MIM process was not triggered. While the product teams collaborated effectively to mitigate the store-level impact, no communications were sent to business stakeholders. To address this, we will now establish a process through service leads to ensure business stakeholders are informed, even when an issue affects only a single store.

On 18/02, when the impact extended to all 150 stores serviced by Thatcham, affecting their stock deliveries, the MIM process was immediately triggered. Relevant communications were sent out, ensuring that business stakeholders remained informed throughout the resolution process.

5.4. Incident handling/ stakeholder management –

The product teams collaborated effectively on 17/02 & 18/02 to mitigate the impact to stores, with no concerns raised regarding the incident handling process. However, the identified gaps in communication will be addressed through a structured process to ensure better transparency in

future incidents.

5.5. Business Communication –

Although there were gaps in communication with wider stakeholders on 17/02, a new process will be established to ensure better incident management in the future.

For the 18/02, there were no concerns from business stakeholders, as regular and timely communications were sent through the service leads – however, a workshop will be conducted with service leads to align the comms in business centric language and keeping the LOOP updates as the single source of truth for any technical actions.

5.6. Supplier Engagement –

There were no suppliers engaged in this incident.

Mitigation Actions:

👉 Agree actions for each of the findings to prevent recurrence, strengthen system reliability & stability, and improve response efficiency for future incidents

	≡ Action	👤 Owner	≡ Due Date	≡ Remarks/ Updates	🏷️ Action Category
1	5.1 Assess the overall observability within GIST to identify significant volumes of stock adjustment messages or inflated allocations at the store level.	James Sherman (GIST)	18th March, 2025	We are currently working with GIST to analyze their monitoring mechanisms and identify areas for improvement, aiming to enhance proactive detection of such issues in the future	Problem Identification & Alerting
2	5.1 Advise GIST to follow the change management process to ensure M&S stakeholders are informed of any capability or smoke tests for new functionalities.	James Sherman (GIST)	3rd March, 2025	James from GIST has confirmed that M&S will be notified of any changes in their environment, including smoke tests. Additionally, an informational change will be recorded in the MNS BMC Helix system for tracking.	Issue handling/ Management

3	5.2 Assess the feasibility of adopting an alternative approach in CSSM to recover the stock position, rather than nullifying adjustments.	⊕ Ledsome, Nancy ⊕ Arjunan, Akash	18th March, 2025	Currently, positive stock adjustments are nullified by reversing them through the creation of additional events with random data to correct the stock position. The teams are now exploring a more efficient and technically optimized approach, while also evaluating the cascading impact on other downstream systems.	Solution & Recovery
4	5.2 Analyze the behavior in RELEX to ensure it is not accepting negative stock-in-transit values, which have resulted in over-allocations.	⊕ White, Matt	18th March, 2025	In Quantum, negative stock-in-transit values were adjusted to zero, ensuring that appropriate allocations were generated for the stores. An EPIC has been raised with RELX, and they are actively working to identify why negative values are being considered and to determine a potential solution.	Permanent fix
5	5.5 There was a gap in communicating the issue on 17/02 to the wider business stakeholders. Therefore, a defined communication process needs to be established through the service leads to ensure stakeholders are informed in cases where MIM is not involved or when the issue impacts a single store, even if the overall impact is minimal.	⊕ Ramasamy, Valliammai ⊕ Eucharista, Lisy	18th March, 2025	The service leads, in collaboration with the service management team, will identify a structured approach for communication. This will require product teams to inform service leads of any issues where MIM is not involved, so that stakeholders are kept updated at regular intervals throughout the resolution and recovery process.	Business & Tech Comms

Please refer to [Major_ Significant and Key Incident Tracker 16th O...](#) for progress updates on Mitigations.

Technical Details and Useful Documents: